JULIAN ESTEBAN RINCÓN RODRÍGUEZ

Computer Science and Artificial Intelligence | Data Science · Data Analytics · ML/AI Engineering · Software Development
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PROFESSIONAL SUMMARY

Final-year (9th of 9 semesters) Computer Science and Artificial Intelligence student who works across the whole data cycle: I clean and analyze data with SQL and Python, build PySpark pipelines on AWS, train ML models validated with cross-validation and tested once on held-out data, and ship them as tested services and LLM agents. Experience delivering software to clients and working in teams of three to five. Seeking a six-month internship in data science, data analytics, ML/AI engineering or software development for January – June 2027.

EDUCATION

Universidad Sergio Arboleda — Bogotá, Colombia | 2022 – Present
B.Sc. Computer Science and Artificial Intelligence | 9th of 9 semesters | GPA 4.04/5.0, with the last three semesters above 4.5/5.0.

Relevant coursework: Big Data and Data Engineering, Machine Learning, Neural Networks, Natural Language Processing, Modeling and Simulation.

SELECTED PROJECTS

SAVI — Real Estate Valuation Agent (Ames, Iowa) | 2026 | v1–v2 team of 4, v3 individual
Python · XGBoost · LightGBM · PyTorch (DQN) · AWS Lambda · EC2 · SageMaker · RAG

- Audited the team dataset, found that 93% of it was assessor records padded with constant columns, and rebuilt it from public sources: the De Cock sales (2,930) joined to 18,078 parcels, with FHFA and Zillow price indices.
- Reached R^2(log) = 0.921 and 7.8% MAPE with XGBoost in 5-fold cross-validation (LightGBM: 0.912, 8.3%), and trained RL policies that approve, review, or reject valuations; on a held-out test of 378 sales none beat Value Iteration significantly (95% bootstrap interval), and I report it that way.
- Shipped it as an event-driven AWS pipeline (S3 → Lambda → EC2 → SageMaker Spot) at under US$0.05 per run, plus a public web agent (deterministic planner, optional Claude/Groq mode) whose figures all trace to tool outputs on 10 of 10 fixed evaluation questions.

Sentiment Analysis on Sentiment140 with MLflow | Sep. 2026 | Team of 5
Python · scikit-learn · TF-IDF · MLflow Model Registry · FastAPI · AWS EC2 · SageMaker

- Compared preprocessing, representation, and classifier stages under cross-validation against a dummy baseline, with every run tracked in MLflow; picked TF-IDF (uni+bigrams) with logistic regression after an ablation study.
- Retrained the champion on 1.36M tweets, evaluated it once on test (macro-F1 0.824), registered it in the MLflow Model Registry, and served it through a FastAPI endpoint on EC2 with error analysis.

ShopStream — AWS Big Data Pipeline | May – Jul. 2026
PySpark · SQL · AWS Lambda · EMR · Glue · RDS PostgreSQL · Flask

- Built an end-to-end pipeline for 2.5M e-commerce events: S3 ingestion with Lambda validation and quarantine, EMR/PySpark analytics for six KPIs with anomaly detection, Glue ETL, and a PostgreSQL star schema; 86% test coverage and CI/CD.

NEXUS — Personal AI Assistant | 2025 – Present
Python · MCP · SQLite FTS5 · systemd · GCP

- Re-architected a ~47,000-line v1 into a ~8,800-line Python layer with 1,400+ tests: daily briefing, email-to-calendar agenda, and memory shared by three AI agents over MCP, plus a copilot that repairs test-breaking commits in an isolated git worktree and merges only after review.

Additional projects: NYC Taxi on AWS EMR (94M trips, partitioned Athena table queried with SQL); Ayuda Terremoto, an offline-first humanitarian platform (FastAPI, PostgreSQL, React, 80+ tests); Phantom, a multi-agent coding workspace that routes tasks by measured model success (Rust, TypeScript); Victor, a voice race engineer for Linux (219 tests); an NLP API with spaCy and FastAPI (93/100 in a live evaluation).

PROFESSIONAL EXPERIENCE

Full-Stack and AI Developer (Freelance) — Zafra Asesores Tributarios | Girardot, Colombia | Jun. 2026

- Replaced a tax advisory firm's Excel workflow with a sales CRM: migrated its records (81 clients, 85 tax filings) into PostgreSQL through a FastAPI/SQLAlchemy backend with 30+ REST endpoints and a React/Vite prospect pipeline.
- Added 8 LLM features (Groq, Llama 3.3 70B) that summarize client history, draft emails and WhatsApp reminders, and score leads, plus a WhatsApp Cloud API bot that qualifies prospects against DIAN filing thresholds and adds them to the pipeline; handed over with technical documentation.

Full-Stack Developer — ReushiGo | Bogotá, Colombia | May 2026

- Built an ordering platform for a family coffee distributor: a React/Vite admin panel for orders and inventory, an Express/Prisma API with 14 endpoints behind an admin key, and a Telegram sales bot over the real catalog (19 products in 4 categories); 12 automated tests and CI on GitHub Actions.

TECHNICAL SKILLS

Languages: Python (intermediate-advanced), SQL, C, Bash; TypeScript, JavaScript and Rust (basic)

Data analysis: Pandas, NumPy, Matplotlib, Power BI, Jupyter, SQL (PostgreSQL, Athena), bootstrap confidence intervals, cross-validation

ML and AI: scikit-learn, XGBoost, LightGBM, PyTorch, spaCy, MLflow (Model Registry); LLM agents with tool use (Claude API, Groq), RAG, MCP, agent evaluation

Data engineering and cloud: PySpark, AWS (S3, EMR, Glue, Athena, RDS, Lambda, EC2, SageMaker), GCP, Terraform, Docker (basic)

Software development: FastAPI, React/Vite, Express/Prisma, PostgreSQL, Git, GitHub Actions, pytest, Linux (Fedora as daily OS)

ACHIEVEMENTS

1st Place — Wisibilízalas International Competition, Universitat Pompeu Fabra | 2019

- Technical product manager and frontend/UX lead of the winning web application, representing my school from Ecuador in the Senior category at 17, against teams from Latin America and Europe.

1st Place — USABOT Robotics Challenge | November 2025

- Led the CAD design and sensor architecture of a smart irrigation system and delivered the pitch, winning against university teams.

CERTIFICATIONS

AWS Academy Graduate — Data Engineering | Amazon Web Services Academy | May 2026 (40 hours)

Database Programming with SQL; Database Design | Oracle Academy | 2024

LANGUAGES

Spanish: native. English: certified B2 overall (iTEP Academic-Plus, November 2025; Listening C1).